

August 15 Gaussian integers revisited



We have seen that $\mathbb{R}[x] \bmod x^2+1$ is

really the same as $\mathbb{C} := \mathbb{R}[i]$

Similarly, we may consider $\mathbb{Z}_p[i]$ and

ask is it true that $\mathbb{Z}_p[i]$ the same

as $\mathbb{Z}_p[x] \bmod x^2+1$.

Lemma 1: $\mathbb{Z}_p[x] \bmod x^2+1 \cong \mathbb{Z}_p[i]$

as commutative rings.

proof: Consider $\rho: \mathbb{Z}_p[x] \bmod x^2+1 \rightarrow \mathbb{Z}_p[i]$.

by $\rho(f(x)) = ev_i(f) = f(i)$. This is a well-defined

ring homomorphism. Note that if

$[f(x)] = [f'(x)]$, then $f'(x) - f(x) = (x^2+1)g(x)$

for some $g(x) \in \mathbb{Z}_p[x]$. and hence

$$\begin{aligned} f(i) &= ev_i(f(x) + (x^2+1)g(x)) \\ &= f(i) + (i^2+1)g(i) = f(i). \end{aligned}$$

It is surjective for $\forall a+bi \in \mathbb{Z}_p[i]$.

we have $\rho([bx+a]) = a+bi$

and it is injective for if $f(i) = 0$,

wlog. assume $f(x) = ax+b$ for $a, b \in \mathbb{Z}_p$.

then $ai+b = 0$ iff $a=b=0$

iff $[f(x)] = [0]$.

But is $\mathbb{Z}_p[i]$ a field?

Recall the lemma that when K is a field.

$K[x] \bmod f(x)$ is a field iff $f(x)$ is

irreducible in $K[x]$. So it implies

$\mathbb{Z}_p[i] \simeq \mathbb{Z}_p[x] \bmod x^2+1$ is a field

iff x^2+1 is irreducible in $\mathbb{Z}_p[x]$.

(note x^2+1 is of deg 2)

iff x^2+1 has a root in $\mathbb{Z}_p$

i.e. $\exists x \in \mathbb{Z}_p$ s.t. $x^2 \equiv -1$.

So we are back to congruence of integers!

Lemma 2 = Let p be an odd prime.

(1) If $p \equiv 1 \pmod{4}$, then $x^2 \equiv -1 \pmod{p}$ is solvable.

(2) If $p \equiv 3 \pmod{4}$, then $x^2 \equiv -1 \pmod{p}$ is not solvable.

proof. Note an odd prime p is either

$p \equiv 1 \pmod{4}$ or $p \equiv 3 \pmod{4}$.

We show (2) by contrapositive. That is.

if $x^2 \equiv -1 \pmod{p}$ is solvable,

then $p \equiv 1 \pmod{4}$.

Suppose $\exists x \in \mathbb{Z}_p = \{1, 2, \dots, p-1\}$.

and recall $x^{p-1} \equiv 1 \pmod{p}$.

Now that p is odd and $p-1$ is even

So $\frac{p-1}{2}$ is an integer.

Then $(-1)^{\frac{p-1}{2}} = (x^2)^{\frac{p-1}{2}} = x^{p-1} \equiv 1 \pmod{p}$

Here $(-1)^{\frac{p-1}{2}} = \begin{cases} +1 & \text{if } \frac{p-1}{2} \text{ is even} \\ -1 & \text{if } \frac{p-1}{2} \text{ is odd} \end{cases}$

Then $\frac{p-1}{2}$ is even implies $p \equiv 1 \pmod{4}$.

Next we show (1) directly. First we

need a result called the Wilson's lemma:

for p an odd prime. $(p-1)! \equiv -1 \pmod{p}$

Now note $1 \times 2 \times 3 \times \dots \times (p-1)$

$$= 1 \times 2 \times \dots \times \frac{p-1}{2} \times (p - \frac{p-1}{2}) \times \dots \times (p-1)$$

$$= 1 \times 2 \times \dots \times \frac{p-1}{2} \times (-\frac{p-1}{2}) \times \dots \times (-1)$$

$$= (-1)^{\frac{p-1}{2}} \left(1 \times 2 \times \dots \times \frac{p-1}{2} \right)^2 \equiv -1 \pmod{p}$$

if $p \equiv 1 \pmod{4}$, then $\frac{p-1}{2}$ is even.

In particular $\left(1 \times 2 \times \dots \times \frac{p-1}{2} \right)^2 \equiv -1 \pmod{p}$

This solves $x^2 \equiv -1 \pmod{p}$.

Corollary 3: let p be an odd prime.

then $\mathbb{Z}_p[i] \simeq \mathbb{Z}_p[x] \bmod x^2+1$ is a field

iff x^2+1 is irreducible in $\mathbb{Z}_p[x]$

iff $x^2 \equiv -1 \bmod p$ is not solvable

iff $p \equiv 3 \bmod 4$.

We now give another characterization on

when $\mathbb{Z}_p[i]$ is a field.

Lemma 4: let p be a prime in $\mathbb{Z}$.

$\mathbb{Z}_p[i]$ is a field iff p is irreducible
in $\mathbb{Z}[i]$.

proof: Consider $\rho: \mathbb{Z}[i] \rightarrow \mathbb{Z}_p[i]$

sending $a+bi$ to $\bar{a}+\bar{b}i$

which is a surjective ring homomorphism.

Note $\bar{a}+\bar{b}i = 0$ in $\mathbb{Z}_p[i]$

iff $a \equiv 0 \pmod{p}$, $b \equiv 0 \pmod{p}$

iff $p \mid a$, $p \mid b$ iff $a+bi \in (p)$

So $\ker \rho = (p)$. By the first isomorphism

theorem $\mathbb{Z}_p[i] \cong \mathbb{Z}[i]/(p)$.

Since $\mathbb{Z}[i]$ is ED, PID, and UFD.

$\mathbb{Z}[i]/(p)$ is a field iff

(p) is maximal ideal iff

p is irreducible in $\mathbb{Z}[i]$.

Corollary 5. Let p be an odd prime. in $\mathbb{Z}$

p is irreducible in $\mathbb{Z}[i]$

iff $x^2 \equiv -1 \pmod{p}$ is not solvable in $\mathbb{Z}_p$

iff $p \equiv 3 \pmod{4}$.

eg. $5 = (2+i)(2-i)$ in $\mathbb{Z}[i]$.

$2+i$ is irreducible in $\mathbb{Z}[i]$.